

TITLE: Vision-based Anomaly detection

TEAM NAME: CLOVER

TEAM ID: YS305

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DOMAIN: AI&ML

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PROBLEM STATEMENT

- **Manual inspection of manufacturing products is slow and inconsistent. Small defects such as cracks, scratches, and dents may go unnoticed due to variations in lighting, product orientation, and surface texture, leading to poor quality and increased rework costs.**
- **Develop a vision-based system using OpenCV and deep learning to distinguish normal and defective products, classify defect types, and locate affected regions accurately under varying image conditions while minimizing false detections and missed defects.**

PROPOSED SOLUTION

- **Automated Pipeline:** Replaces manual checks with an end-to-end intelligent computer vision workflow.
- **Core Flow:** Product Image → OpenCV Preprocessing → AI Detection → Defect Localization → Classification → PASS/FAIL Decision.
- **Value Proposition:** Fast, consistent, and data-driven quality control minimizing missed defects and operational downtime.

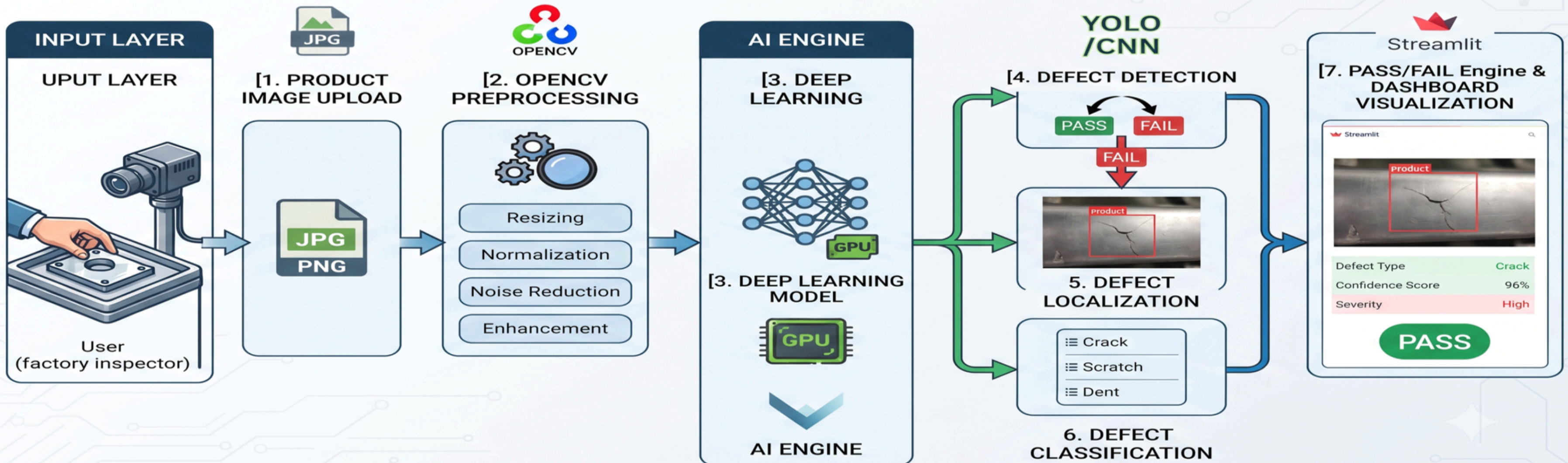
OBJECTIVE

- **Automate high-speed quality checks to eliminate human fatigue and handle scale.**
- **Accurately detect and localize micro-defects (cracks, scratches, dents, holes) prior to customer delivery.**
- **Ensure robustness against varying lighting conditions, backgrounds, and product orientations.**
- **Maximize model Recall to completely prevent defective products from passing.**
- **Deliver actionable real-time outputs via an intuitive Streamlit dashboard**

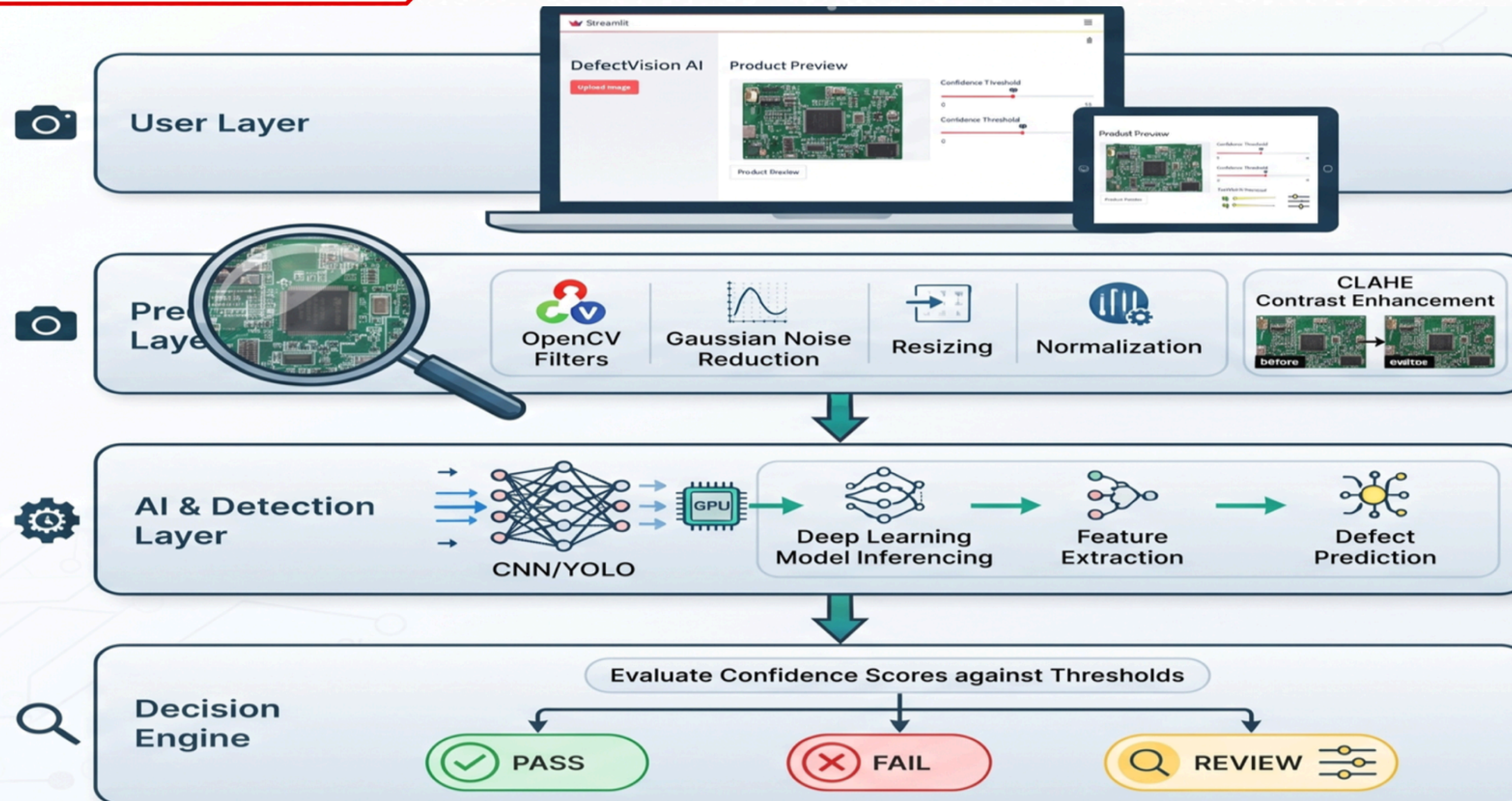
METHODOLOGY

- **Data Acquisition & Labeling: Gathering and structuring industrial product images categorized by defect classes.**
- **Transfer Learning: Fine-tuning pre-trained CNN/YOLO architectures specifically for manufacturing surface defect identification.**
- **Pipeline Integration: Bridging OpenCV preprocessing modules with deep learning inference scripts inside an interactive dashboard**

WORKFLOW



SYSTEM ARCHITECTURE



IMPLEMENTATION

- **Frontend: Streamlit**
- **Backend & Logic: Python, Pandas, NumPy**
- **Computer Vision & AI: OpenCV, YOLO / CNN-based transfer learning models**
- **Evaluation Metrics: Precision, Recall, F1-score, mAP, and Confusion Matrix**



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*Thank
you*

